NAME: MUHAMMAD HAMZAH IQBAL

REGISTRATION NUMBER: F24604018

DEPARTMENT: COMPUTER ENGINEERING

SUBJECT: DISCRETE STRUCTURES

ASSIGNMENT: 1

INSTRUCTOR: DR MUHAMMAD SAJJAD

Q1 $A = \{+, -, \times, \div\}$

$R_1 = A \times A$

$R_1 = \{(+,+), (+,-), (+,\times), (+,\div), (-,+), (-,-), (-,\times), (-,\div), (\times,+), (\times,-), (\times,\times), (\times,\div), (\div,+), (\div,-), (\div,\times), (\div,\div)\}$

1. Equivalence Relation (Reflective, Symmetric, Transitive)

Reflective	Symmetric	Transitive
• R_1 is a reflective set as $(x,x) \in R$ for all $x \in R_1$. • All elements present have their reflection ~~set~~ elements present	• R_1 is [redacted] symmetric as for (x,y) there is (y,x) present.	• R_1 is also transitive because for (x,y) (y,z) there is (x,z) also present.

→ Therefore since all properties of Equivalence relation is satisfied R_1 is an Equivilance relation.

2. Partially Ordered Set (Reflective, Antisymmetric, Transitive)

Reflective	Antisymmetric	Transitive
• R_1 is reflective as for all (x,x) there is a reflective element present	• R_1 is not antisymmetric as for every (a,b) there is (b,a)	• R_1 is transitive as for (x,y) and (y,z) there is also (x,z)

→ Because R_1 is not ~~sy~~ antisymmetric therefore it is not a partially ordered set.

R_1 is also a function because for each element set A in R_1 there is a unique element present in set B of R_1.

$$R_2 = \{(+,+),(-,-),(\times,\times),(\div,\div)\}$$

1. Equilance Relation

Reflective	Symmetric	Transitive
It is reflective because $(x, x) \in R_2$ for all element in R_2	It is symmetric because R_2 contains (a, b) for every (b, a).	It is transitive because R_2 has (x, y), (y, z) and (x, z).

→ R_2 is an Equivalence relation as all three properties are satisfied.

2. POSET (Partially ordered set)

Reflective	Antisymmetric	Transitive
It is reflective because (x, x) is present for all possible pairs.	It is antisymmetric because all pairs are equal and no (a, b) to (b, a) condition needs to be checked.	It is transitive because no different pair elements are present.

R_2 is also a Partially ordered set.

3. Funtion :- It is a funtion because for every element of set A there is a unique element of set B in each pair.

$R_3 = \{(+,+), (+,-), (-,-), (\times,\times), (\div,\div)$

1. Equivalence Relation

Reflective	Symmetric	Transitive
It is reflective because each element has its copy present. For every x there is a subsequent x present.	It is not symmetric because for $(+,-)$ there is not $(-,+)$ present.	It is transitive because no (a,b) (b,c) and (a,c) is present so transitive property doesn't not need to be checked.

- Since Symmetric property is not satisfied it is not a equivalence relation.

2. POSET

Reflective	Antisymmetric	Transitive
It is reflective because each element has its copy present	It is antisymmetric $(+,-)$ has no $(-,+)$ element	It is transitive because for every (a,b) (b,c) there is (a,c) present.

* Since all three conditions have been satisfied it (R_3) is a Partially ordered set.

3. Function

R_3 is not a function because for every element of set A there is not a unique element of set B because of $\underline{(+,+)}$ and $\underline{(+,-)}$.

$R_4 = \{ (+,+), (+,-), (-,-), (-,+), (\times,\times), (\div,\div) \}$

1. Equivalence Set

Reflective	Symmetric	Transitive
R_4 is reflective because for each element x there is a copy present (x, x)	R_4 is symmetric because for every (a, b) there is (b, a) present.	R_4 is transitive because for every (a, b), (b, c) there is (a, c) present.

→ Since all three properties are satisfied R_4 is a equivalence set.

2. POSET

Reflective	Antisymmetric	Transitive
R_4 is reflective because for each element x there is a copy present	R_4 is not antisymmetric because for every (a, b) there is (b, a) present.	R_4 is transitive because for every (a, b), (b, c) there is (a, c) present.

→ Since Antisymmetric property is not satisfied R_4 is not a POSET.

3. Function

R_4 is not a function because for each element of set A there is not a unique element of set B present. $\underbrace{(+,+),(+,-)}$

∴ Not a function.

$R_5 = \{(+,+), (-,-), (+,-), (\times,\times), (-,\times), (\div,\div)\}$

1. Equivalence Relation

Reflective	Symmetric	Transitive
R_5 is reflective because for every element x there is a copy present (x, x)	R_5 is not symmetric because for $(-,\times)$ there is no $(\times,-)$ present.	R_5 is not transitive because for every (x,y), (y,z) there is (x,z) present

→ Since Symmetric property and transitive property is not satisfied R_5 is not Equivalence relation.

2. POSET

Reflective	Antisymmetric	Transitive
R_5 is reflective because for every element x there is a copy present. (x, x)	R_5 is antisymmetric because for (a,b) there is no (b,a) present, $(-,\times)$.	R_5 is not transitive because for $(+,-)$ $(-,\times)$ there is not $(+,\times)$ present.

→ R_5 is not POSET because transitive property is not satisfied.

3. Function

R_5 is not a function because for every element of set A there is not a unique element of set B present. $(-,-)$, $(+,-)$, $(-,\times)$, $(+,+)$.

Q2

(a) Given that $A \subseteq B$, $B \cap C \neq \phi$ and $A \cap C = \phi$ verify distributive law.

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

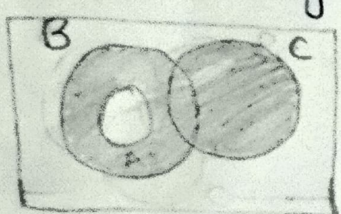
- $A \subseteq B$ implies $A \cap B = A$
- $A \cap C = \phi$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

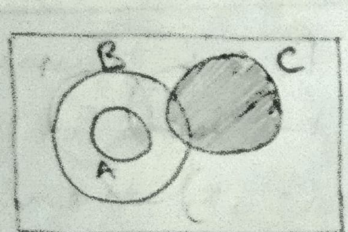
$$A \cap (B \cup C) = A \cup \phi = A$$

Both sides are equal therefore it is true now lets check through Venn diagram.

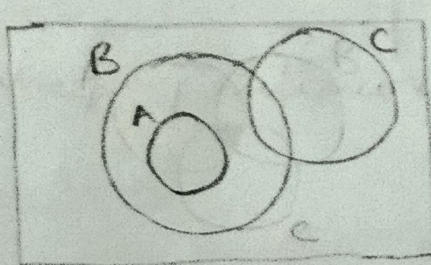
RHS Venn Diagram



$B \cup C$

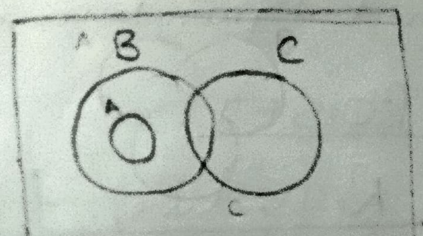


C

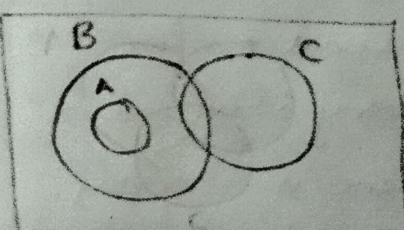


$A \cap (B \cup C)$

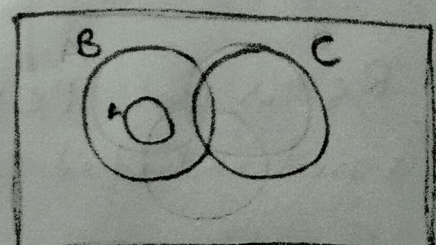
LHS Venn diagram



$A \cap B$



$A \cap C$



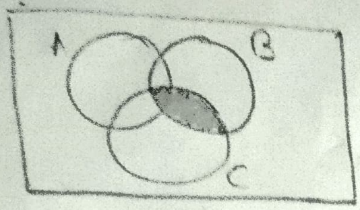
$A \cap B \cup A \cap C$

Since Right hand side and left hand side are equal so this law has been proved

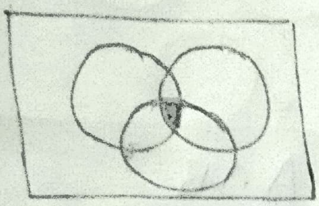
(b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

LHS

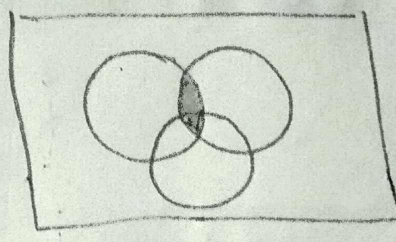
RHS



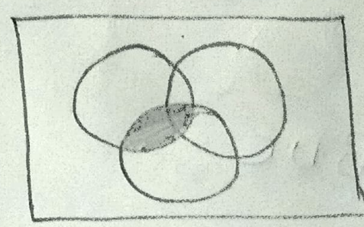
$B \cap C$



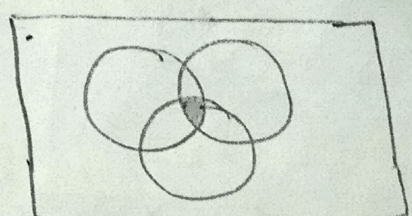
$A \cup (B \cap C)$



$A \cup B$



$A \cup C$



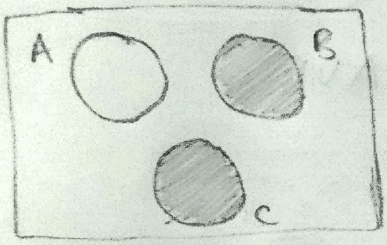
$(A \cup B) \cap (A \cup C)$

Since LHS is equal to RHS therefore this is proved through this Venn diagram

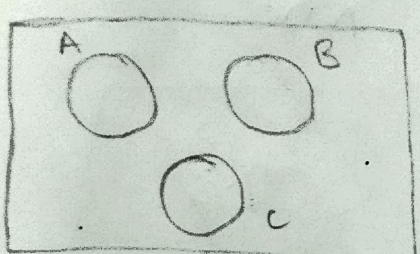
(c) If A, B and C are disjoint sets, verify the distributive law

$$A \cap (B \cup C) = A \cap B \cup A \cap C$$

LHS

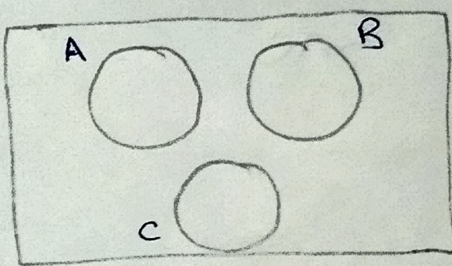


$B \cup C$

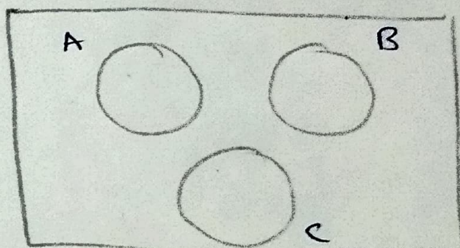


$A \cap (B \cup C)$

RHS



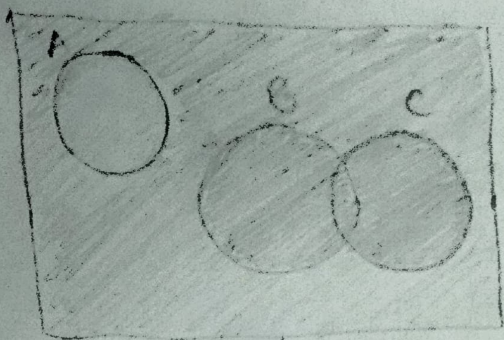
$A \cap B$



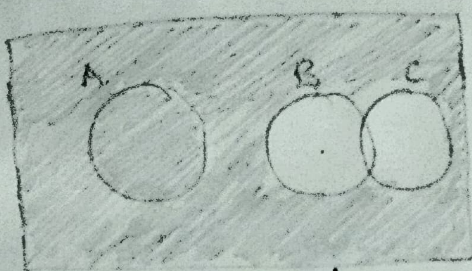
$A \cap B \cup A \cap C$

Since Right hand side is equal to Left hand side hence it is proved

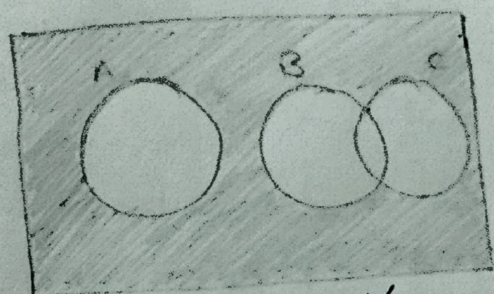
d) If A and B are disjoint A and C are disjoint but B and C are overlap, draw the venn Diagram for the expresion



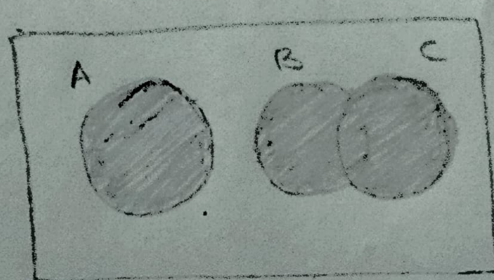
A'



$(B \cup C)'$



$A' \cap (B \cup C)'$



$(A' \cap (B \cup C)')'$